

Global Environmental Politics

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Previous class: introduction to cooperation

- Climate change has diffuse, long-term impacts:
 - Global impacts, but disproportionate in developing countries.
 - May induce scarcity: ↓ agricultural yields + ↑ climate migration.
 - ... leading to conflict over resources.

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 - **Collective action problems undermine cooperation.**
 - Some (powerful) stakeholders may oppose/hamstring regulation.

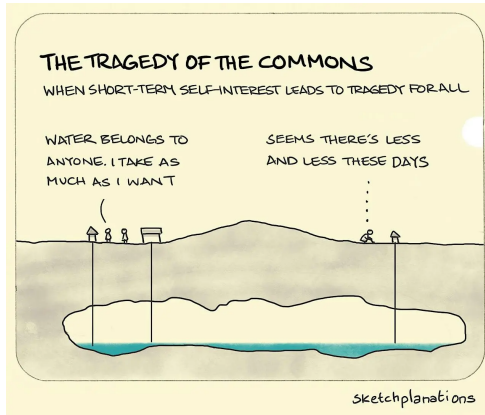
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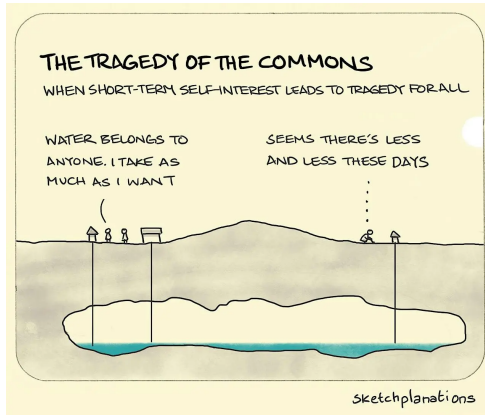
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- Phasing out CFCs was less costly; lengthy; hard to replicate.
 - ... also imperfect: HFCs ⇒ green house effect.

Tragedy of the commons



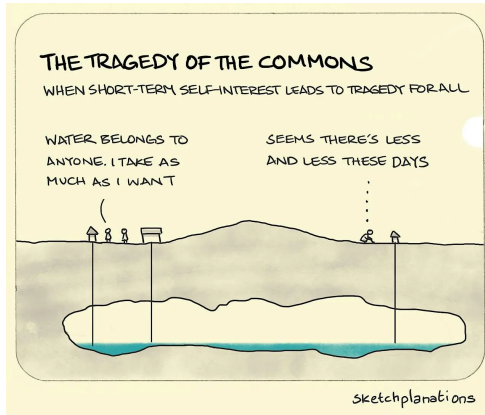
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- Creates a *negative externality*.
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Contracts and their limits

- Party A *pays* party B a compensation for the negative externality.
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 - E.g., local or national government.
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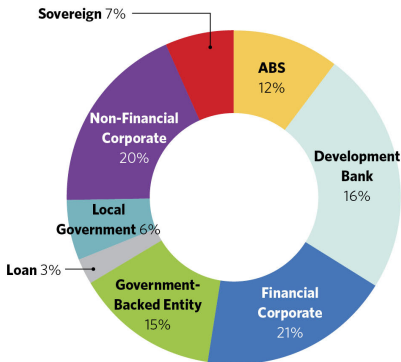
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 - Recall, there's not really a third party that can enforce contracts.

Contracts and their limits

The \$1 trillion: green bond issuers



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■ Green bonds are an example but they have not been very effective.

■ Conditions are not always enforced, binding; incentives as ESGs

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The prisoners dilemma and no cooperation

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A	Cooperate	1,1	-2,2
	No cooperate	2,-2	-1,-1

- Two actors: A and B
- Two actions: cooperate, not cooperate
- A's (B's) pay-offs depend on his action, and that of B (A).
- We seek for a *Nash equilibrium*...

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- The outcome is that which makes them worse off!

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$$\sum_{t=0} \delta^t 1 > 2 + \sum_{t=1} \delta^t (-1);$$

$$\frac{1}{1-\delta} > 2 - \frac{\delta}{1-\delta};$$

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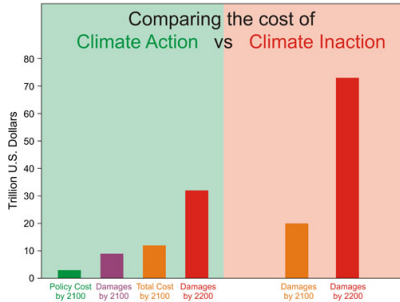
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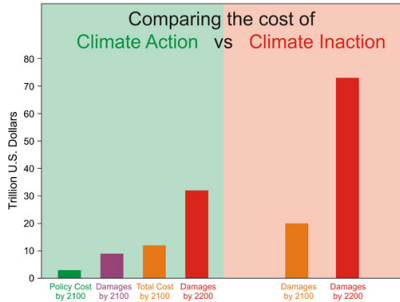
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But there is evidence for inter-temporal inconsistency!



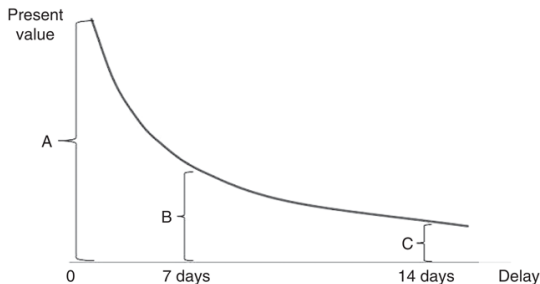
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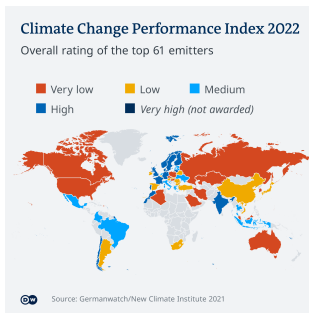
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Source: Authors' own elaboration

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Assume $p(n) \in [0, 1]$ falls with group size (n):

$$\sum_{t=0} \delta^t [p(n-1) - 2(1-p(n-1))] > \\ ([2 \cdot p(n-1) - 1(1-p(n-1))] + \sum_{t=1} \delta^t [-1 + 2p(n-1)/t]; \\ \frac{p(n-1) - 1}{1 - \delta} > \frac{-\delta}{1 - \delta}; \\ \delta > 1 - p(n-1).$$

- Likelihood of no cooperation increases with group size!
- Harder to punish when there are many (heterogenous) actors!
 - High cost of conflict, sanctions, etc., encourage cooperation. (why?)
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- Compensation, redistribution becomes relevant!
- To sustain cooperation we need to redistribute the gains, compensating the losers!